

COMP30271 Cognitive Computing

Coursework presentation

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Team: 3

#	Functionality/Behaviour	Summarise the work done. Provide file names and line numbers where the majority of the code is implemented (but not code snippets!)	Target mark [0..3]
1	Basic Mapping: Create simple occupancy grid using ground-truth Gazebo poses in simulation.	Completed by Arjan	3
2	Basic Navigation: Implement tele-op navigation in Gazebo simulation.	Completed by Arjan	3
3	Object Detection: Basic detection and classification of objects (signals, poster-like landmarks) in simulation.	Completed by Arjan	3
4	Goal Position Detection: Implement detection of goal markers using object detection system in simulation.	Completed by Yasmine	3
5	Traffic Rules: Basic traffic sign recognition and navigation response in simulation.	Completed by Abdullah	3
6	Enhance Navigation: Implementation of Nav2-based navigation including obstacles avoidance and path optimization in simulation.	Completed by Yasmine	3

7	Landmark Database: Store and manage detected landmark information.	Completed by Me & Akshay, I have contributed by creating an in-depth word document on how to accomplish this function. Akshay implemented the steps + code into the system. Files and lines numbers have not been told to me by Akshay, Akshay in the one who implemented the code onto the system.	3
8	Advanced Object Detection: Counting objects (i.e. oranges, trees, vehicles) in poster-like landmarks in simulation.	Completed by Me & Akshay, I have contributed by creating an in-depth word document on how to accomplish this function. Akshay implemented the steps + code into the system. Files and lines numbers have not been told to me by Akshay, Akshay in the one who implemented the code onto the system.	3
9	Visual Odometry: Implementation of basic VO system for real robot localization.	Completed by Akshay & Arjan, Rakhshan, Abdullah and Yasmine Present. Before beginning I gave my word document guide on how to do functionality 9 to Akshay which he looked through and improved then implemented onto the system.	3
10	Real Robot SLAM and navigation	Not completed	3